higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

FITTING AND TURNING
NQF LEVEL 2

22 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1

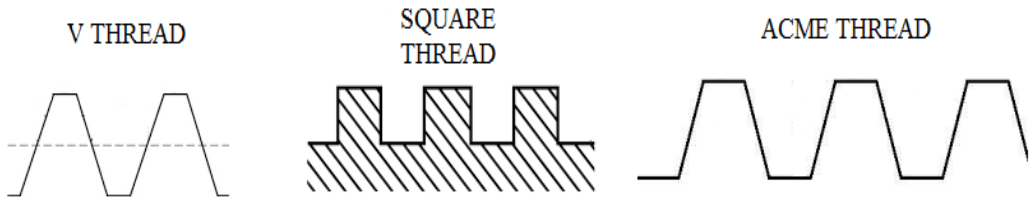
1.1
1.1.1 A
1.1.2 B
1.1.3 B
1.1.4 A
1.1.5 A

(5× 2) (10)

1.2
- The ring test must be done before installing the wheel on the grinder.
- Begin by suspending the wheel with a string through the centre. For smaller, lighter wheels place your finger through the centre hole/bore.
- Use a non-metal tool and tap the stone at every 45°.
- If there is no crack on the wheel, the sound will be sharp and continuous.
- A cracked wheel will have a dull sound. Do not assemble the stone.

(Any other relevant answers) (5)

1.3



Labelling = 1 mark
Sketch = 1 mark (6)

1.4
- Machine reamer
- Parallel reamer
- Taper reamer
- Helical flute reamer
- Expanding reamer
- Adjustable reamer

(Any 4 × 1) (4)
[25]

QUESTION 2

2.1
A Pulley
B Motor
C Pillar
D Base
E Table
F Chuck
G Hand-feed lever
H Machine guard (8)

2.2
2.2.1 D
2.2.2 E
2.2.3 A
2.2.4 F
2.2.5 G
2.2.6 B
2.2.7 C
2.2.8 H

(8 × 1) (8)

2.3

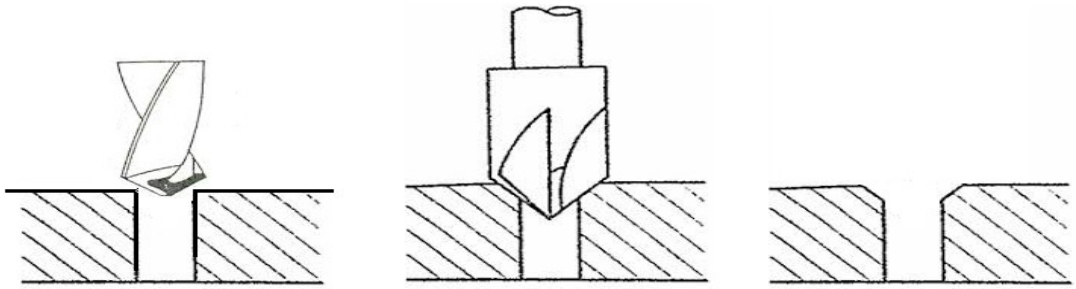


FIGURE 1 FIGURE 2 FIGURE 3

FIGURE 1 = 1 mark✓
Drill the work piece with a steel drill bit, and remove the drill bit from the chuck. ✓

FIGURE 2 = 1 mark.✓
Insert a counter-sink drill into the chuck, and drill the work piece to the height of the head of the screw. ✓

FIGURE 3 = 1 mark.✓
Completed hole to accommodate a countersunk screw.✓
(Any other relevant answers 3 × 2) (6)

2.4
- Drill bit not sharpened correctly.
- Incorrect feed.
- Drill bit clogged with metal chips.
- Insufficient coolant.
- Incorrect speed. (Any other relevant answers 3 × 1) (3)

[25]

QUESTION 3

3.1
3.1.1 True
3.1.2 True
3.1.3 True
3.1.4 False
3.1.5 True

(5 × 1) (5)

3.2
- Plain turning
- Facing
- Taper turning
- Screw cutting
- Parting off
- Drilling
- Boring (Any other relevant answers 5 × 1) (5)

3.3
S = ?
N = 159,15 RPM
D = 60 mm divided 1 000 = 0,06 m

S = π × D × N✓
= π × 0,06 × 159,15✓✓
= 30,003 m/min. ✓✓
= 30 m/min (5)

3.4 The completed dimension of the work piece is 10 mm. However, a deviation in the size is allowed.✓
The acceptable size can be from 10 mm to 0,02 mm = 9,98 mm✓✓ to 10 mm + 0,02 mm = 10,02 mm.✓✓ (5)

3.5
- Check the dimensions of the work piece.
- Check for clamping marks on the work piece.
- Check for sharp edges.
- Check the overall length.
- Check for correct finish. (Any other relevant answers) (5)

[25]

QUESTION 4

4.1
- Ensure that the correct personal protective equipment is worn.
- Ensure that the machine guards are in place.
- Ensure that the work area is free and spacious.
- Make sure that the milling machine is off when placing or removing work pieces.
- Use hand protection or gloves when installing milling cutters.
- Use guards or shields to deflect steel chips.
- Ensure that the work area is clean and free of steel chips, spills and fluid.

(Any other relevant answers 5 × 1) (5)

4.2
- Helical milling cutter
- Side-and-face cutter
- Dovetail cutter
- T-slot cutter
- Corner-rounding cutter
- End mill
- Slot drill
- Slitting cutter
- Convex cutter
- Concave cutter (Any other relevant answers 5 × 1) (5)

4.3
4.3.1 milling operations
4.3..2 cutting speed
4.3..3 material
4.3..4 cutting tool
4.3..5 clamping arrangements

(5 × 1) (5)

4.4
- Consider the feed rate.
- The condition of the machine.
- The type of material being machined.
- Condition of the material.
- The type of cutting tool. (5)

4.5
- Prevents the formation of continuous chips.
- Assists in the disposal of material being cut.
- Ensures a better cutting action.
- A good finish is acquired.
- Chattering is reduced.
- Less power consumption. (Any other relevant answers 5 × 1) (5)

[25]

TOTAL: 100

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